

GEOGRAPHY

PHYSICAL GEOGRAPHY



GEOMORPHOLOGY

MODULE - 20

VOLCANOES

- **PLATE BOUNDARY INTERACTIONS**

VOLCANOES

2014 GS-1 MAINS QUESTION

Q 1 Why are the world's fold mountain systems located along the margins of continents? Bring out the association between the global distribution of Fold Mountains and the earthquakes and volcanoes. **(10 Marks) (150 words)**

HINTS:

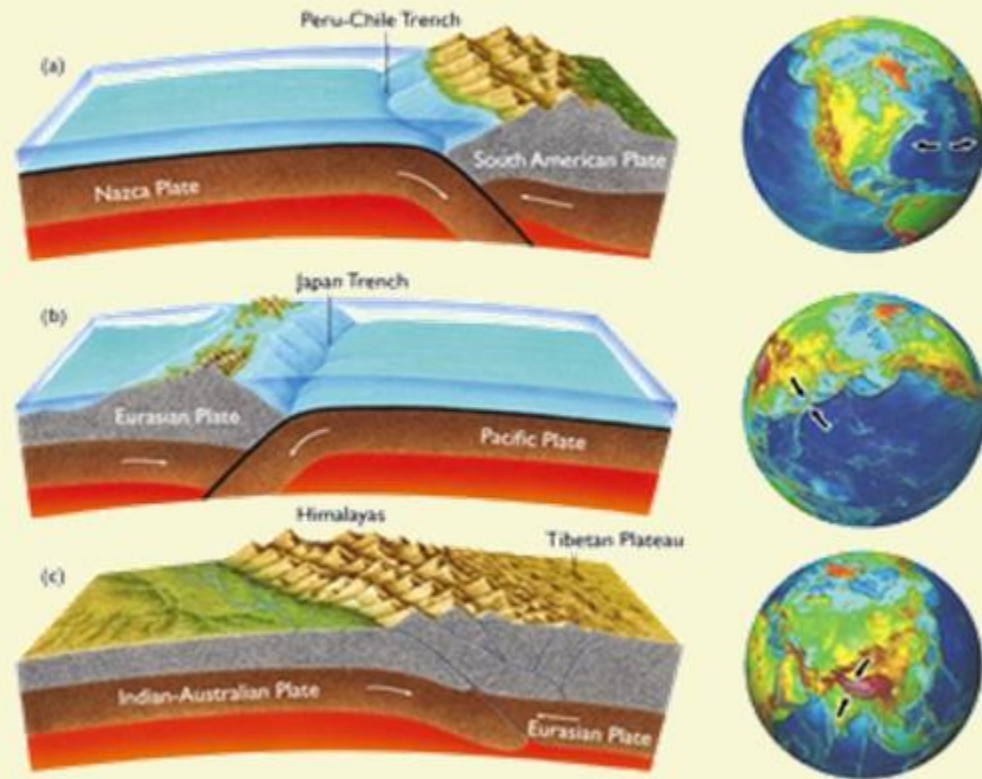
1. Link Plate Tectonics concept and Mountain Building or Orogenesis.
2. Think about convergent plate boundaries collision.
3. Volcanoes – At Himalayas or At Andes?

Q 2 Explain the formation of thousands of islands in Indonesian and Philippines archipelagos. **(10 Marks) (200 words)**

HINTS:

1. Ocean-ocean Convergence also known as island-arc convergence.
2. Also look at the “Mechanism of Vulcanism” part in this module.

CONVERGENT BOUNDARY INTERACTIONS



Depending on type of Lithospheric plates involved in the convergent process, there can be **three possibilities of boundary interactions**.

1. Ocean-Ocean convergence
2. Ocean continent convergence
3. Continent-Continent Convergence

Convergent Boundaries (Subduction)

CONVERGENT BOUNDARY INTERACTIONS

OCEAN-OCEAN CONVERGENCE



1 Depending on the **relative size of the oceanic plates** one of the limb or plate will get subducted into the mantle.

2 After reaching depth of about 100 kms, the **subducted plate melts and magma rises upwards**. Due to enormous pressure and continuous upward movement of the magma, it creates **volcanic explosions at the ocean floor**.

CONVERGENT BOUNDARY INTERACTIONS

OCEAN-OCEAN CONVERGENCE



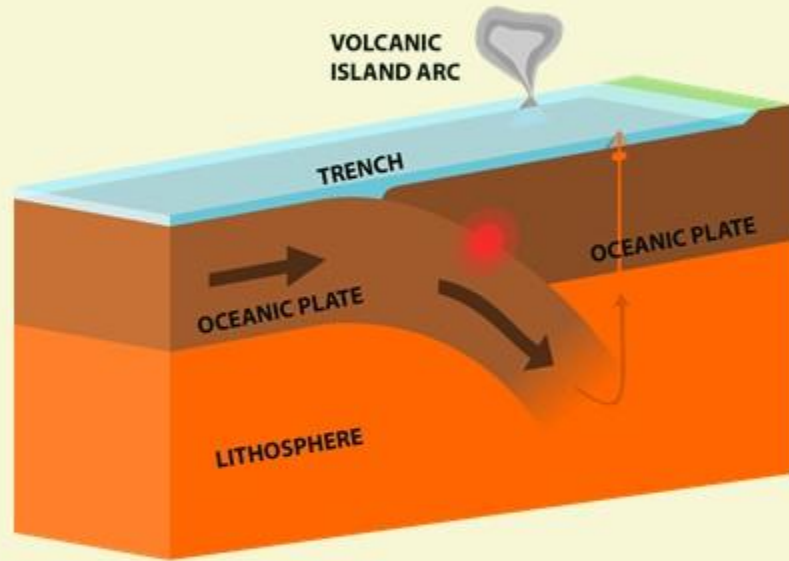
3 Constant volcanism above the subduction zone creates layers of rocks.

4 As this process continues for millions of years, a volcanic landform is created which in some cases rises above the ocean waters.

5 Such volcanic landforms when they are high enough they form islands or island arcs which is also called as Archipelagos. Example, the Indonesian Archipelago, Philippines, Japanese and many of the Caribbean islands.

CONVERGENT BOUNDARY INTERACTIONS

OCEAN-OCEAN CONVERGENCE



WHAT DO YOU MEAN BY TRENCHES?

Trenches are narrow topographic depressions which define the natural boundaries between two lithospheric plates. They are also the deepest parts of the ocean floor.

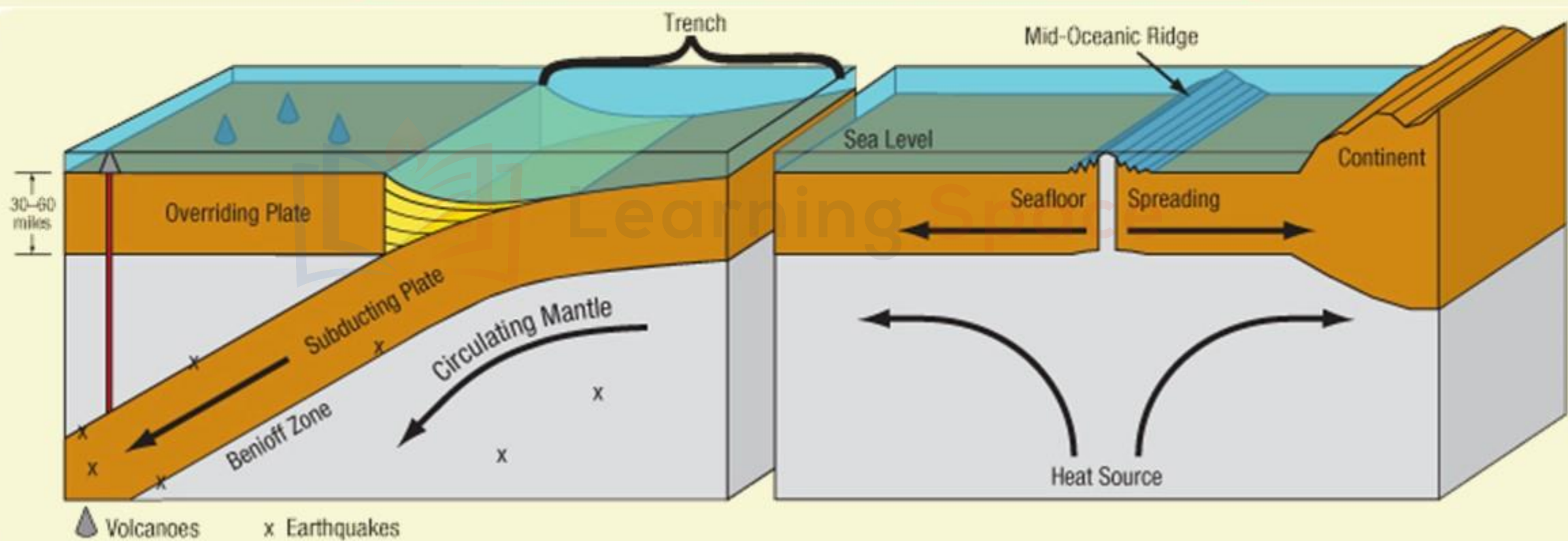
6 Where one slab is getting subducted beneath another **a trench gets formed**. This is the reason for the presence of many trenches like Japanese Trench, Mariana Trench, Philippines Trench, Kuril Trench in the Pacific Ocean. These trenches **indicate subduction of oceanic plates in the Pacific Ring of Fire**.

7 Thus, Ocean-Ocean convergence is otherwise known as **Island Arc Convergence** because it results in the formation of island arcs.

CONVERGENT BOUNDARY INTERACTIONS

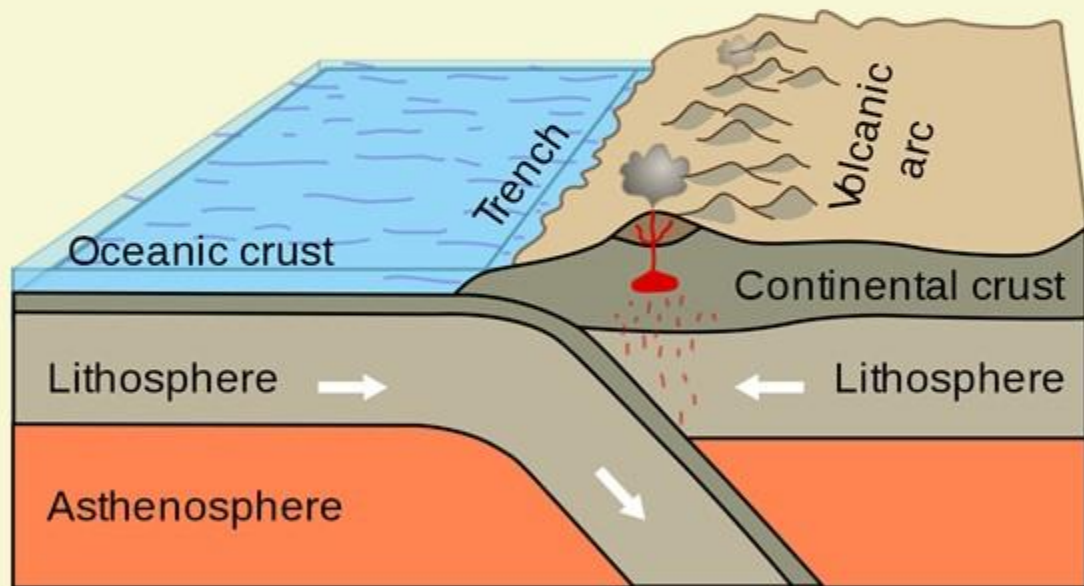
OCEAN-OCEAN CONVERGENCE

TRENCH IN THE SUBDUCTION ZONE



CONVERGENT BOUNDARY INTERACTIONS

OCEAN-CONTINENT CONVERGENCE

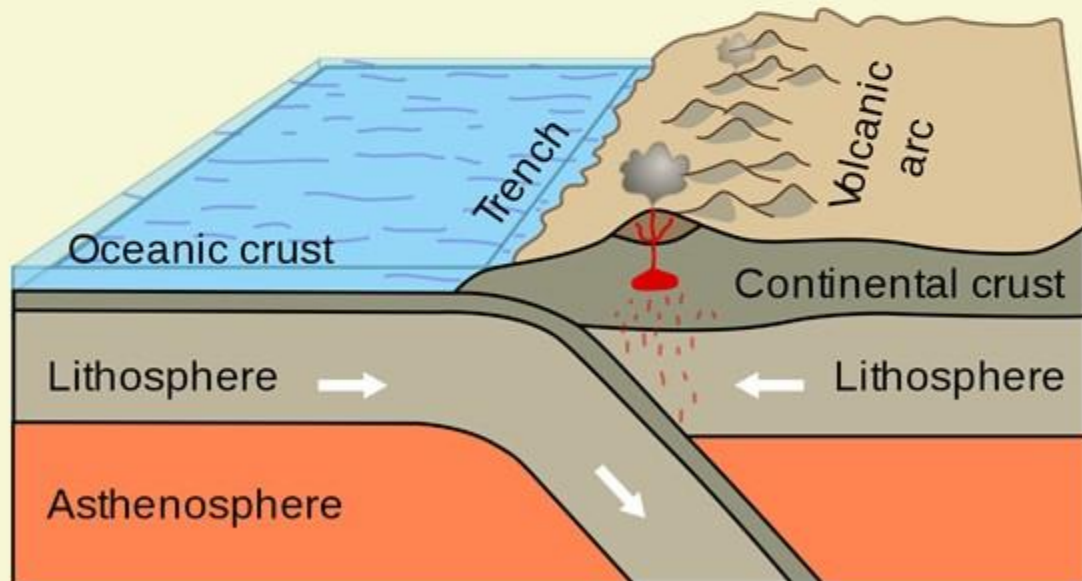


- 1 O-C Convergence is similar to O-O Convergence but **instead of formation of island arcs it would result in the formation of volcanic arcs.**
- 2 When an Oceanic and Continental Plate collide with each other, **the oceanic plate being denser subducts beneath the lighter continental plate forming a trench along the boundary.**
- 3 After reaching a certain depth, the plates melt. Magma, which comprises of **metamorphosed sediments and the melting part of subducting plate** rises upwards and creates constant volcanic eruptions at the surface of the continental plate.

CONVERGENT BOUNDARY INTERACTIONS

OCEAN-CONTINENT CONVERGENCE

4 This continuous eruptions all along the boundary of the continental margins will lead to the **formation of continental arc**, which is nothing but narrow chain of volcanic mountains.



5 Orogenesis happens because of advancing oceanic plate that exerts a compressive stress upon the continental margin and leads to the folding of volcanic accumulations creating a fold mountain system.

6 These volcanic accumulations along the edges of the continent **form fold mountain chains like the Rockies and the Andes**.

CONVERGENT BOUNDARY INTERACTIONS

OCEAN-CONTINENT CONVERGENCE



CORDILLERAS OF ANDES

7 The ranges so formed are also known as **Cordilleras** which is defined as fold mountains present along the edges of **continent-oceanic margin** and are series of parallel ranges.

8 Even in this case because of the subduction of oceanic plate there would be the **creation of trenches** all along the **continent-oceanic margin**.

9 For example, we have the **Chilli Trench**, **Peru Trench**, **Californian Trench** along the American coast and **Andes mountain** is a result of subduction of Nazca plate beneath the South American plate.

CONVERGENT BOUNDARY INTERACTIONS

OCEAN-CONTINENT CONVERGENCE



10 As this Ocean-Continent Convergence is associated with the **subduction of plates** under the continents there will be **volcanoes along the fold mountains**.

11 Example, Ojos Del Salado, Cotopaxi, **Chimborazo** are some of the volcanic mountains present in the Andes mountain range and these are associated with the Circum-Pacific belt of volcanoes.

CONVERGENT BOUNDARY INTERACTIONS

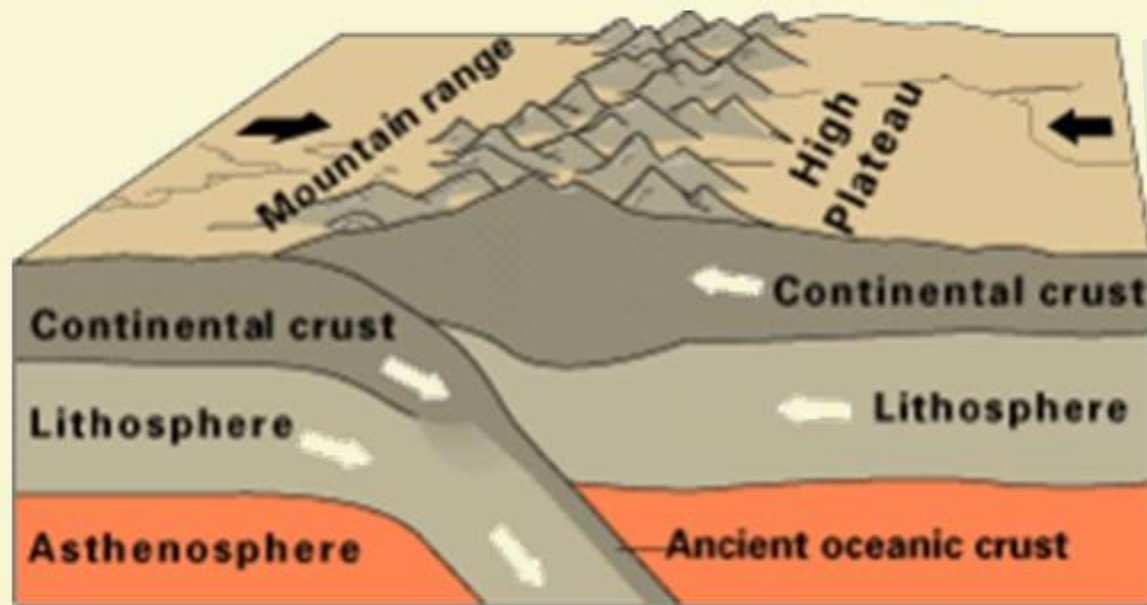
CONTINENT-CONTINENT CONVERGENCE

1 C-C collision may be seen as the last phase in the development of convergent boundary interactions.

2 Here, since both the plates are continental plates **they are light and buoyant and they will resist subduction.**

3 So, in C-C collision the compressive forces are the strongest and also the sediments scraped in between the continents are the maximum.

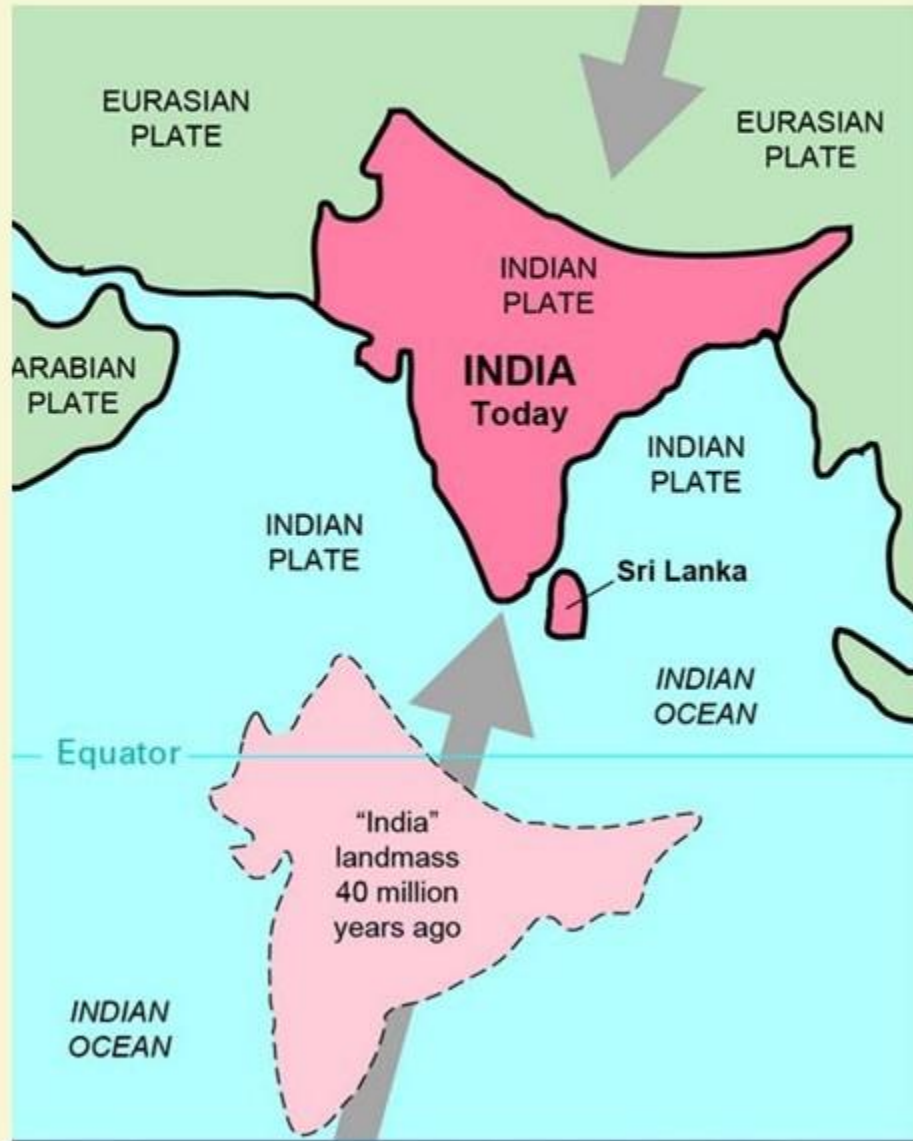
4 In C-C collision, it is possible for one of the continental block **to get under thrust over another continental block.** This adds to the elevation of the mountain systems because of the **doubling of the crust.**



Continental-continental convergence

CONVERGENT BOUNDARY INTERACTIONS

CONTINENT-CONTINENT CONVERGENCE



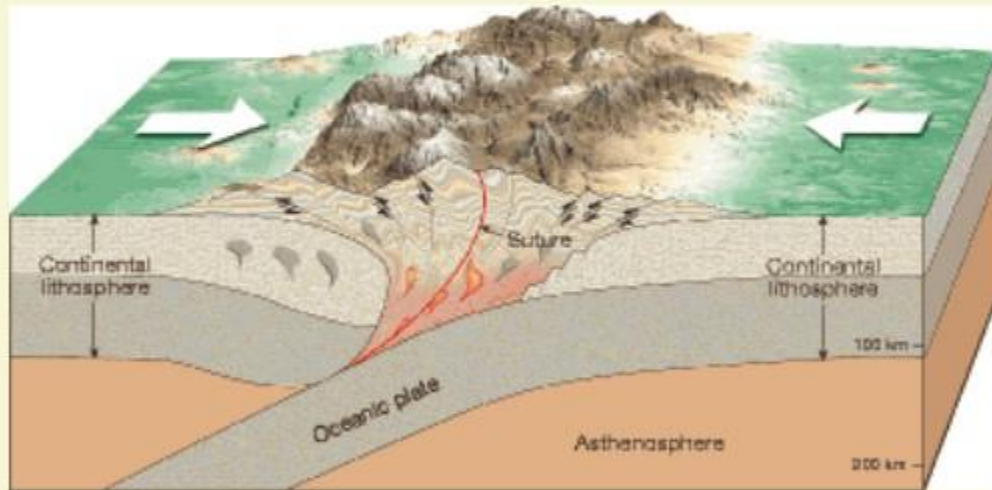
5 The fold mountains that gets formed by this process are compressed and uplifted to huge height.

6 In fact these kind of collisions will result in the highest mountain systems of the world.

7 In South Asia, the Indian plate has under thrust the Eurasian plate, which has elevated the mountains along the Himalayas and thus they form the highest mountain systems of the world.

CONVERGENT BOUNDARY INTERACTIONS

CONTINENT-CONTINENT CONVERGENCE



8 In C-C collision because there is no subduction taking place such boundaries will not have active volcanism but they can have volcanic rocks which are remnants of past volcanic activity.

9 For example, in the northern mountain complex of South Asia Karakoram mountains and Dras hills have such volcanic rocks associated with subduction of ancient Tethys oceanic floor.

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